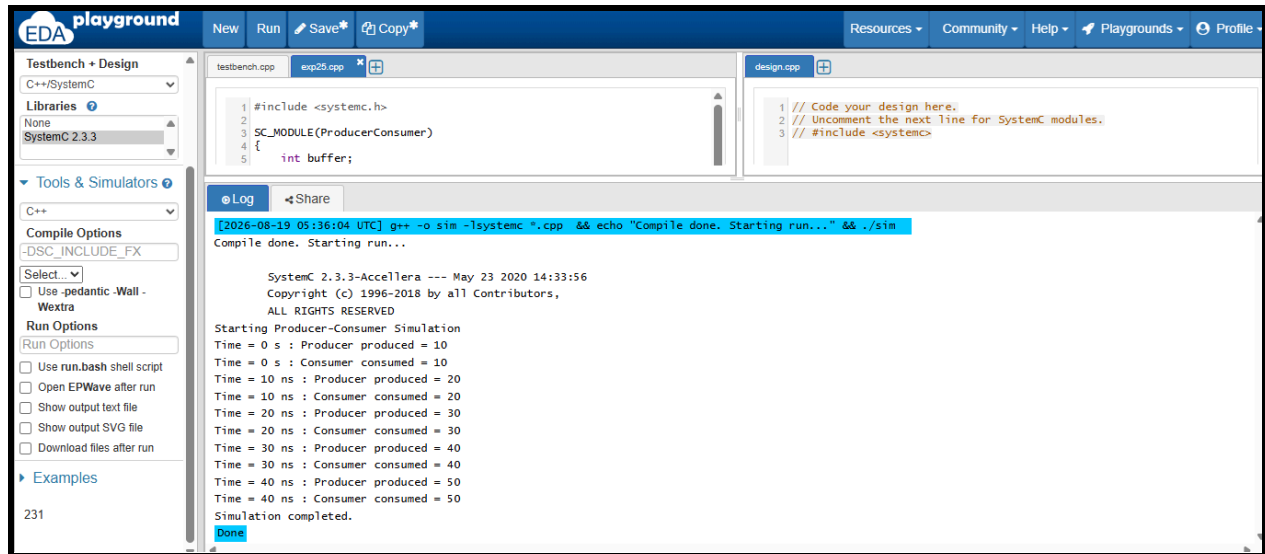




C++ Program:

```
#include <systemc.h>
```

```
SC_MODULE(ProducerConsumer)
```

```
{
```

```
    int buffer;
```

```
    bool buffer_full;
```

```
    void producer()
```

```
    {
```

```
        for(int i = 1; i <= 5; i++)
```

```
        {
```

```
            while(buffer_full)
```

```
                wait(10, SC_NS);
```

```
            buffer = i * 10;
```

```
            buffer_full = true;
```

```
            cout << "Time = " << sc_time_stamp()
```

```
                << " : Producer produced = "
```

```
                << buffer << endl;
```

```
            wait(10, SC_NS);
```

```
        }
```

```
    }
```

```
    void consumer()
```

```
{
```

```

for(int i = 1; i <= 5; i++)
{
    while(!buffer_full)
        wait(10, SC_NS);

    cout << "Time = " << sc_time_stamp()
        << " : Consumer consumed = "
        << buffer << endl;

    buffer_full = false;

    wait(10, SC_NS);
}
}

SC_CTOR(ProducerConsumer)
{
    buffer = 0;
    buffer_full = false;

    SC_THREAD(producer);
    SC_THREAD(consumer);
}

};

int sc_main(int argc, char* argv[])
{
    ProducerConsumer pc("ProducerConsumer");

    cout << "Starting Producer-Consumer Simulation"
        << endl;

    sc_start(120, SC_NS);

    cout << "Simulation completed." << endl;

    return 0;
}

```